

# NT219- Cryptography

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## Week 3: Modern Symmetric Ciphers

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# What is cryptograph?

- Cryptology= Cryptography + Cryptanalysis

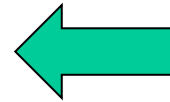
## Goals

- Confidentiality
- Privacy

**What?**

### Cipher systems

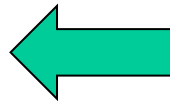
- Symmetric (AES)
- Asymmetric (RSA, ECC, CRYSTALS-KYBER)



- Integrity
- Authentication
- Non-repudiation (Accountability)

### Hash functions

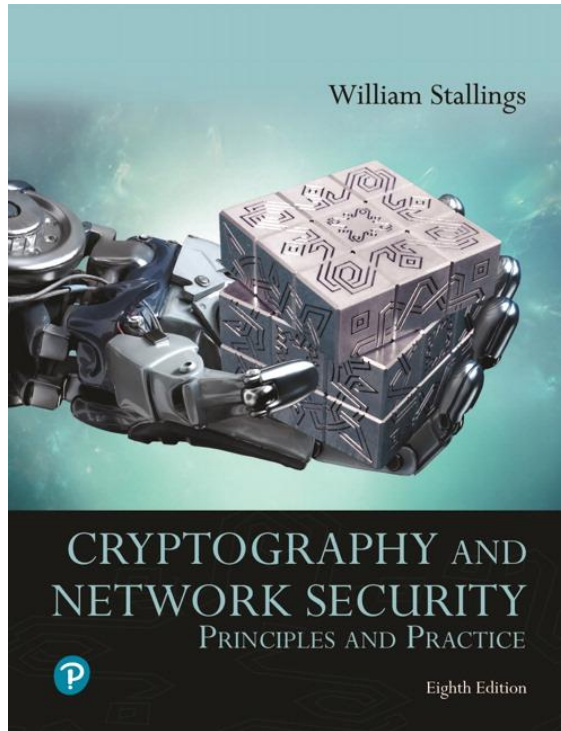
Message authentication code (MAC)  
Digital signature (digital certificate)



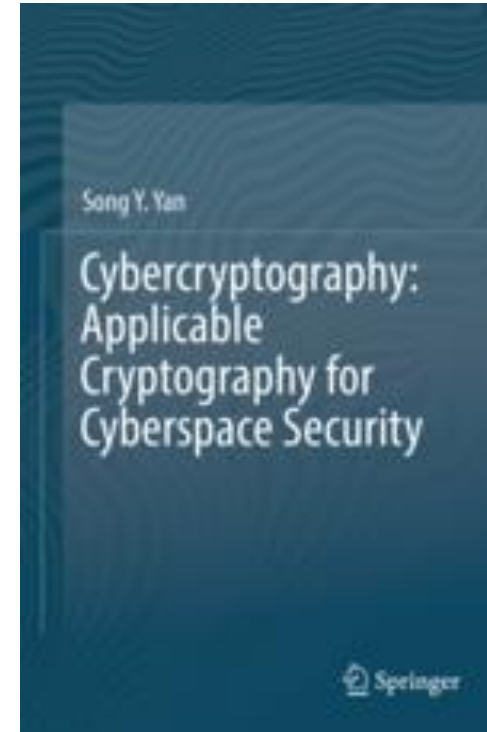
- Availability

# Textbooks and References

## ■ Text books



[1] Chapter 4,6



[2] Chapter 5

# Classical cipher algorithms

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## ■ Substitution Technique

### ➤ Monoalphabetic cipher

- Replace one character by another character
- Replace one character by other characters

### ➤ Polyalphabetic cipher

- Replace some characters by other characters
  - 2 by 2:
  - 3 by 3 or  $n$  by  $n$

## ■ Transposition Technique

- ### ➤ Keep the same source characters but change their positions

# Polyalphabetic Cipher

- Polyalphabetic Cipher: **Replace some characters by other characters**
  - **Playfair Cipher**: replace 2 characters by 2 characters
  - **Hill Cipher**: replace 3 characters by 3 characters, ...
  - **Vigenère Cipher**



# Playfair encryption

Key matrix

|   |   |   |     |   |  |      |
|---|---|---|-----|---|--|------|
|   |   |   |     |   |  | → +1 |
| M | O | N | A   | R |  | ↓ +1 |
| C | H | Y | B   | D |  |      |
| E | F | G | I/J | K |  |      |
| L | P | Q | S   | T |  |      |
| U | V | W | X   | Z |  |      |

**Plaintext:** "Hide the gold in the tree stump"

**Plaintext diagram:**

HI DE TH EG OL DI NT HE TR EX ES TU MP

**Ciphertext diagram:**

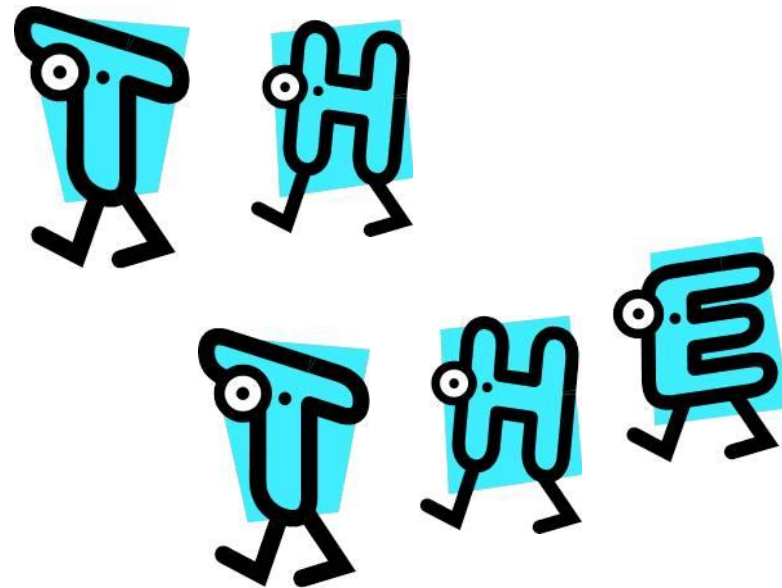
BF CK PD FI .. .. .. ZD

[https://en.wikipedia.org/wiki/Playfair\\_cipher](https://en.wikipedia.org/wiki/Playfair_cipher)

# Polyalphabetic Cipher

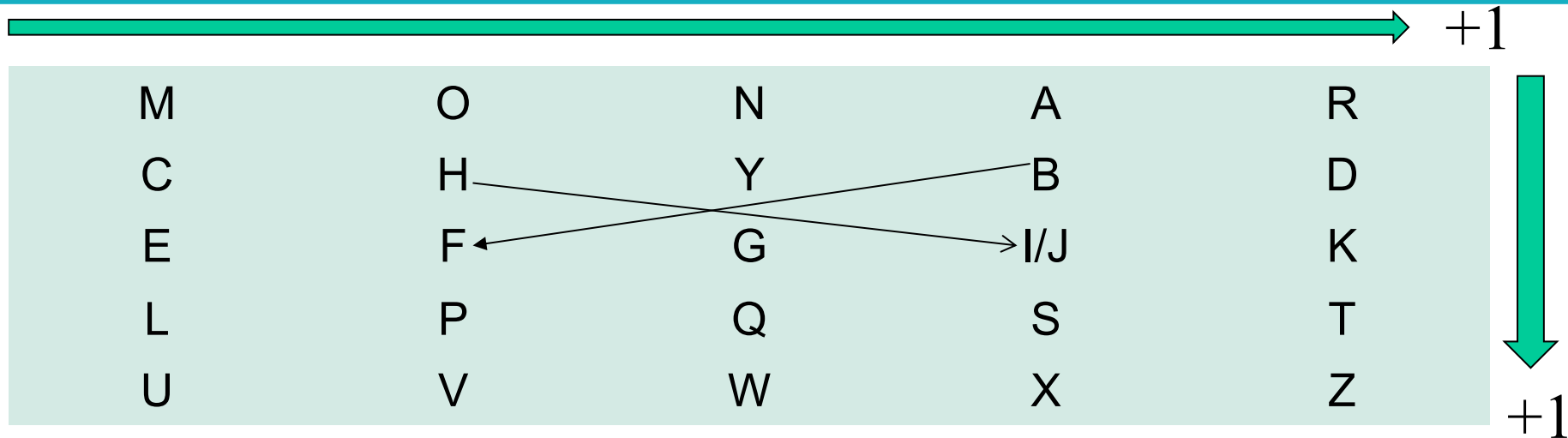
## Cryptoanalysis Playfair cipher

- Digram
  - Two-letter combination
  - Most common is *th*
- Trigram
  - Three-letter combination
  - Most frequent is *the*



# Cryptanalysis Playfair

Guess the  
Key?



|   |   |   |     |   |
|---|---|---|-----|---|
| M | O | N | A   | R |
| C | H | Y | B   | D |
| E | F | G | I/J | K |
| L | P | Q | S   | T |
| U | V | W | X   | Z |

Horizontal shift: +1  
Vertical shift: +1

- **UnigramScorer:** Single letter frequencies;  
[https://en.wikipedia.org/wiki/Frequency\\_analysis](https://en.wikipedia.org/wiki/Frequency_analysis)
- **DigramScorer:** Bigram frequencies  
<https://en.wikipedia.org/wiki/Bigram>
- **QuadgramScorer:** Trigram frequencies  
<https://en.wikipedia.org/wiki/Trigram>

# Polyalphabetic Cipher

## (4) Hill Cipher

|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| A | B | C | D | E | F | G | H | I | J | K  | L  | M  | N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |

- Developed by the mathematician Lester Hill in 1929
- Strength is that it completely hides single-letter frequencies
  - The use of a larger matrix hides more frequency information
  - A 3 x 3 Hill cipher hides not only single-letter but also two-letter frequency information
- Strong against a ciphertext-only attack but easily broken with a known plaintext attack

$$C = K.P \bmod 26 \quad \begin{pmatrix} k_{1,1} & k_{1,2} & k_{1,3} \\ k_{2,1} & k_{2,2} & k_{2,3} \\ k_{3,1} & k_{3,2} & k_{3,3} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \bmod 26$$

## (5) Vigenère Cipher

- Best known and one of the simplest polyalphabetic substitution ciphers
- In this scheme the set of related monoalphabetic substitution rules consists of the 26 Caesar ciphers with shifts of 0 through 25
- Each cipher is denoted by a key letter which is the ciphertext letter that substitutes for the plaintext letter a

[https://en.wikipedia.org/wiki/Vigen%C3%A8re\\_cipher](https://en.wikipedia.org/wiki/Vigen%C3%A8re_cipher)

# Vigenère Cipher

## Vigenère matrix

**Key: deceptiv**

|   | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| B | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| C | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| D | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| E | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| F | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| G | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| H | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| I | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| J | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| K | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| L | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| M | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| N | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| O | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| P | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| Q | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| R | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| S | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| T | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| U | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| V | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| W | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| X | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| Y | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| Z | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |

# Example of Vigenère Cipher

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- To encrypt a message, a key is needed that is as long as the message
- Usually, the key is a repeating keyword
- For example, if the keyword is *deceptive*, the message “we are discovered save yourself” is encrypted as:

key:           deceptivedeceptivedeceptive  
plaintext:   wearediscoveredsaveyourself  
ciphertext: ??

# Vigenère Autokey System

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- Example:

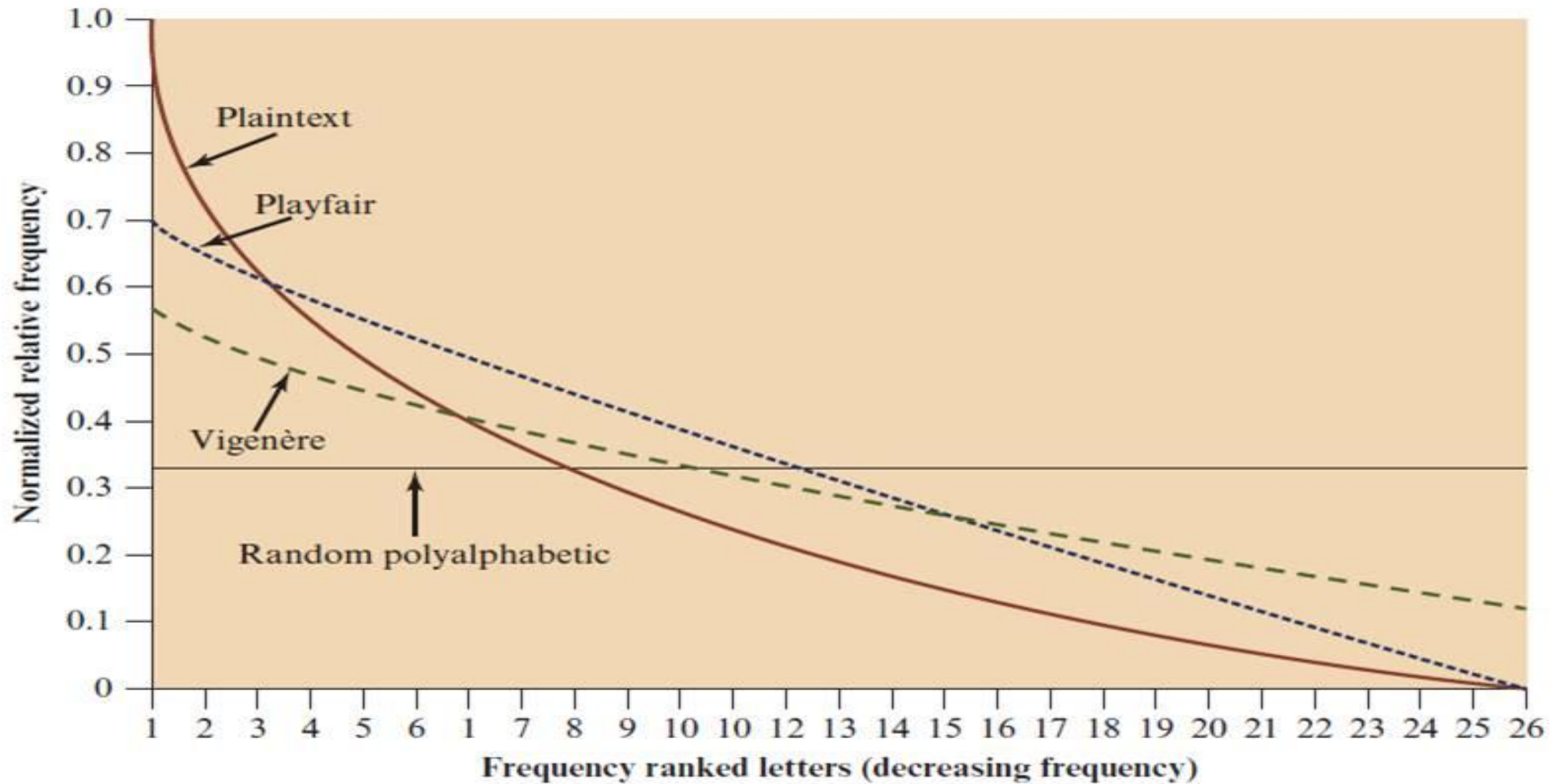
plaintext: w e a r e d i s c o v e r e d s a v e y o u r s e l f

key: d e c e p t I v e w e a r e d i s c o v e r e d s a v

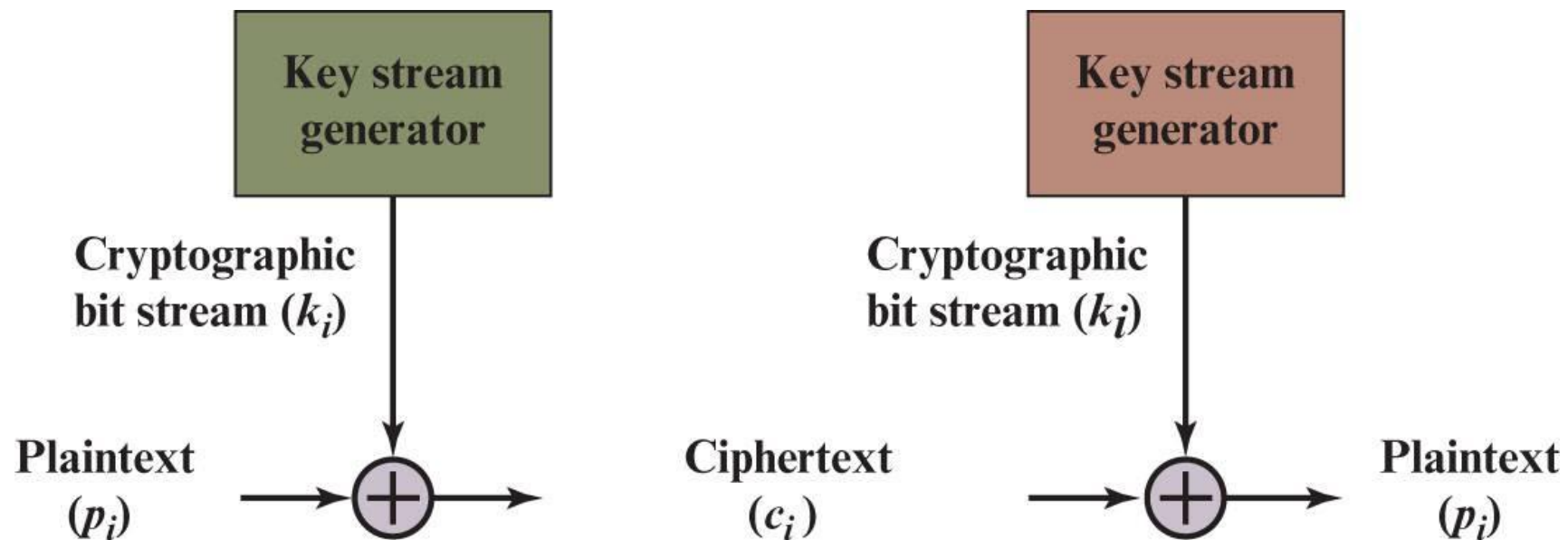
ciphertext: Z I C V T W Q N G K Z E I I G A S X S T S L V V W L A

- Even this scheme is vulnerable to cryptanalysis
  - Because the key and the plaintext share the same frequency distribution of letters, a statistical technique can be applied

# Classical symmetric cipher cryptanalysis



## (6) Vernam Cipher



[https://en.wikipedia.org/wiki/Gilbert\\_Vernam](https://en.wikipedia.org/wiki/Gilbert_Vernam)

# One-Time Pad

- Improvement to Vernam cipher proposed by an Army Signal Corp officer, Joseph Mauborgne
- Use a **random key that is as long as the message** so that the key need not be repeated
- Key is used to encrypt and decrypt a single message and then is discarded
- Each new message requires a new key of the same length as the new message
- **Scheme is unbreakable**
  - Produces random output that bears no statistical relationship to the plaintext
  - Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the code



- The one-time pad offers complete security but, in practice, has two fundamental difficulties:
  - There is the practical problem of making large quantities of random keys
    - Any heavily used system might require millions of random characters on a regular basis
  - Mammoth key distribution problem
    - For every message to be sent, a key of equal length is needed by both sender and receiver
- Because of these difficulties, the one-time pad is of limited utility
  - Useful primarily for low-bandwidth channels requiring very high security
- The one-time pad is the only cryptosystem that exhibits *perfect secrecy* (see Appendix F)

# Transposition ciphers (permutation cipher)

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**Goals: scrambles the positions of characters**

- (1) Rail fence cipher**
- (2) Columnar Transposition Cipher**
- (3)**

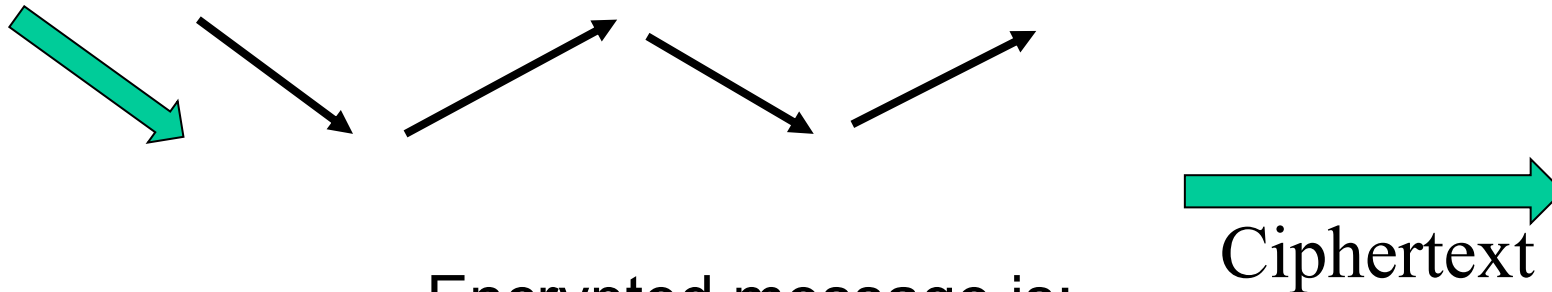


[https://en.wikipedia.org/wiki/Transposition\\_cipher](https://en.wikipedia.org/wiki/Transposition_cipher)

# Transposition cipher

## (1) Rail fence cipher

- Simplest transposition cipher
- Plaintext is written down as a sequence of diagonals and then read off as a sequence of rows
- To encipher the message “**meet me after the toga party**” with a rail fence of depth 2, we would write:



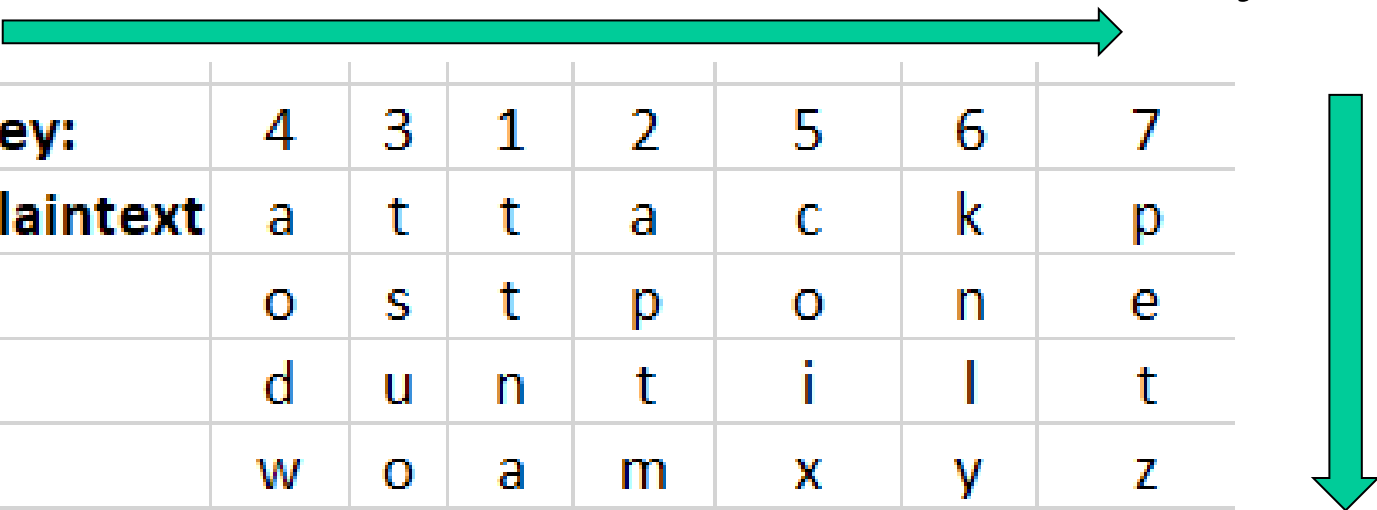
MEMATRHTGPRYETEFETEOAAT

[https://en.wikipedia.org/wiki/Rail\\_fence\\_cipher](https://en.wikipedia.org/wiki/Rail_fence_cipher)



# Columnar Transposition Cipher

- Is a more complex transposition
- Write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns
  - The order of the columns then becomes the key to the algorithm

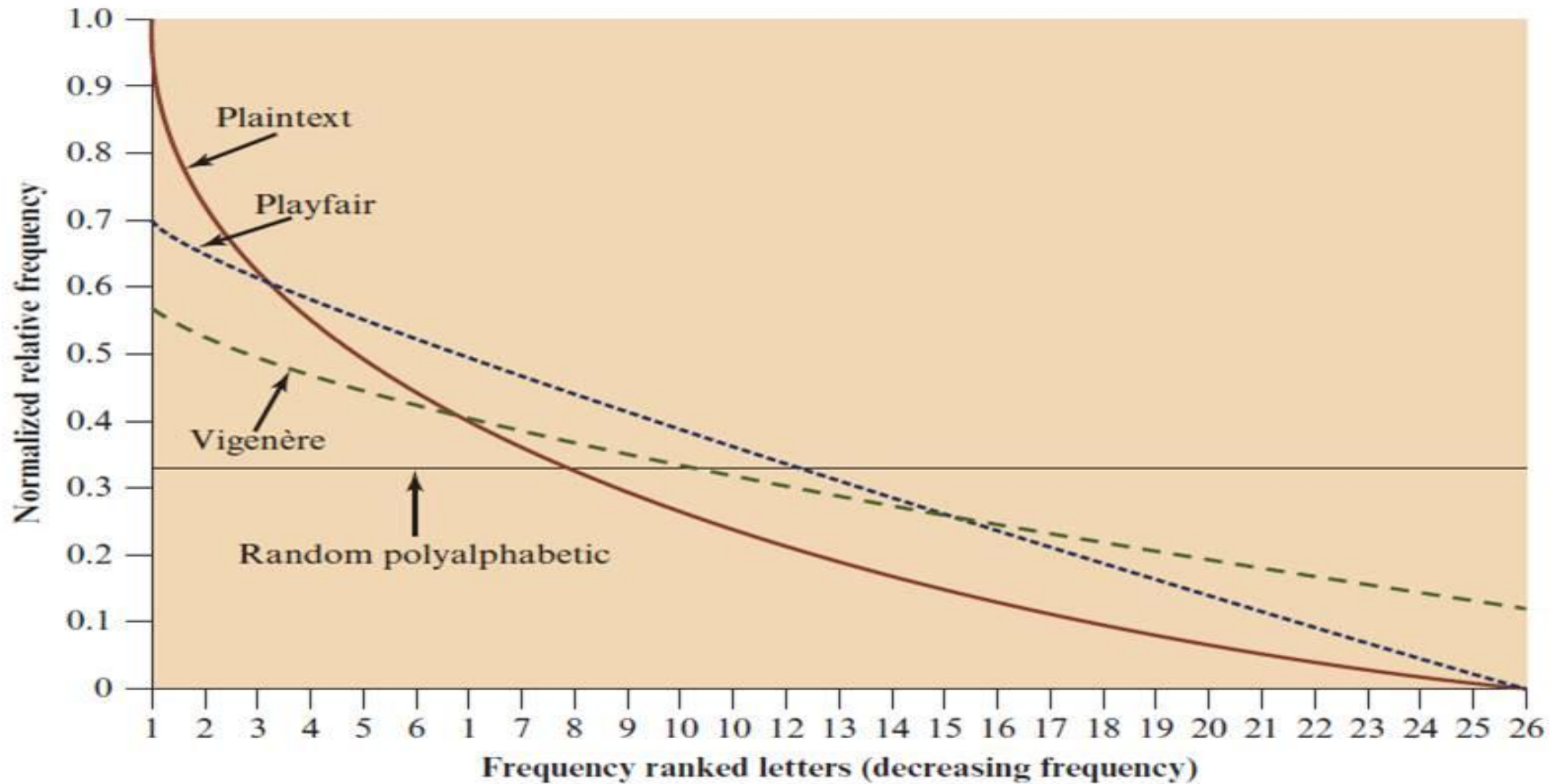


|                  |   |   |   |   |   |   |   |
|------------------|---|---|---|---|---|---|---|
| <b>Key:</b>      | 4 | 3 | 1 | 2 | 5 | 6 | 7 |
| <b>Plaintext</b> | a | t | t | a | c | k | p |
|                  | o | s | t | p | o | n | e |
|                  | d | u | n | t | i | l | t |
|                  | w | o | a | m | x | y | z |
|                  |   |   |   |   |   |   |   |

Ciphertext

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

# Classical symmetric cipher cryptanalysis



| A | B | C | D | E | F | G | H | I | J | K  | L  | M  | N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

### ■ Plaintext stream

|     |   |    |    |   |   |    |   |    |   |   |    |    |
|-----|---|----|----|---|---|----|---|----|---|---|----|----|
| M = | A | T  | T  | A | C | K  | A | T  | D | A | W  | N  |
|     | 0 | 19 | 19 | 0 | 2 | 10 | 0 | 19 | 3 | 0 | 22 | 13 |

### ■ Secret key (Keystream)

|      |    |   |    |    |    |    |   |    |    |    |    |   |
|------|----|---|----|----|----|----|---|----|----|----|----|---|
| K' = | L  | E | M  | O  | N  | L  | E | M  | O  | N  | L  | E |
|      | 11 | 4 | 12 | 14 | 13 | 11 | 4 | 12 | 14 | 13 | 11 | 4 |

### ➤ Ciphertext

|     |    |    |   |    |    |    |   |   |    |    |   |    |
|-----|----|----|---|----|----|----|---|---|----|----|---|----|
| C = | L  | X  | F | O  | P  | V  | E | F | R  | N  | H | R  |
|     | 11 | 23 | 5 | 14 | 15 | 21 | 4 | 5 | 17 | 13 | 7 | 17 |

$$C = c_1 c_2 \cdots c_i \cdots \text{ where } c_i = m_i + k_i \bmod 26$$

# Stream Cipher (2 of 8)

- **Secret key (Keystream)**

$$K = k_1 k_2 \cdots k_i \cdots$$

- **Plaintext stream**

$$M = m_1 m_2 \cdots m_i \cdots$$

$m_i$  : bit or byte

- **Ciphertext**

$$C = c_1 c_2 \cdots c_i \cdots$$

where  $c_i = m_i \oplus \overline{k_i}$

|       |       |       |     |       |
|-------|-------|-------|-----|-------|
| $k_1$ | $k_2$ | $k_3$ | ... | $k_n$ |
|-------|-------|-------|-----|-------|

|       |       |       |     |       |
|-------|-------|-------|-----|-------|
| $m_1$ | $m_2$ | $m_3$ | ... | $m_n$ |
|-------|-------|-------|-----|-------|

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|                  |                  |     |                  |
|------------------|------------------|-----|------------------|
| $k_1 \oplus m_1$ | $k_2 \oplus m_2$ | ... | $k_n \oplus m_n$ |
|------------------|------------------|-----|------------------|

# Stream Cipher (3 of 8)

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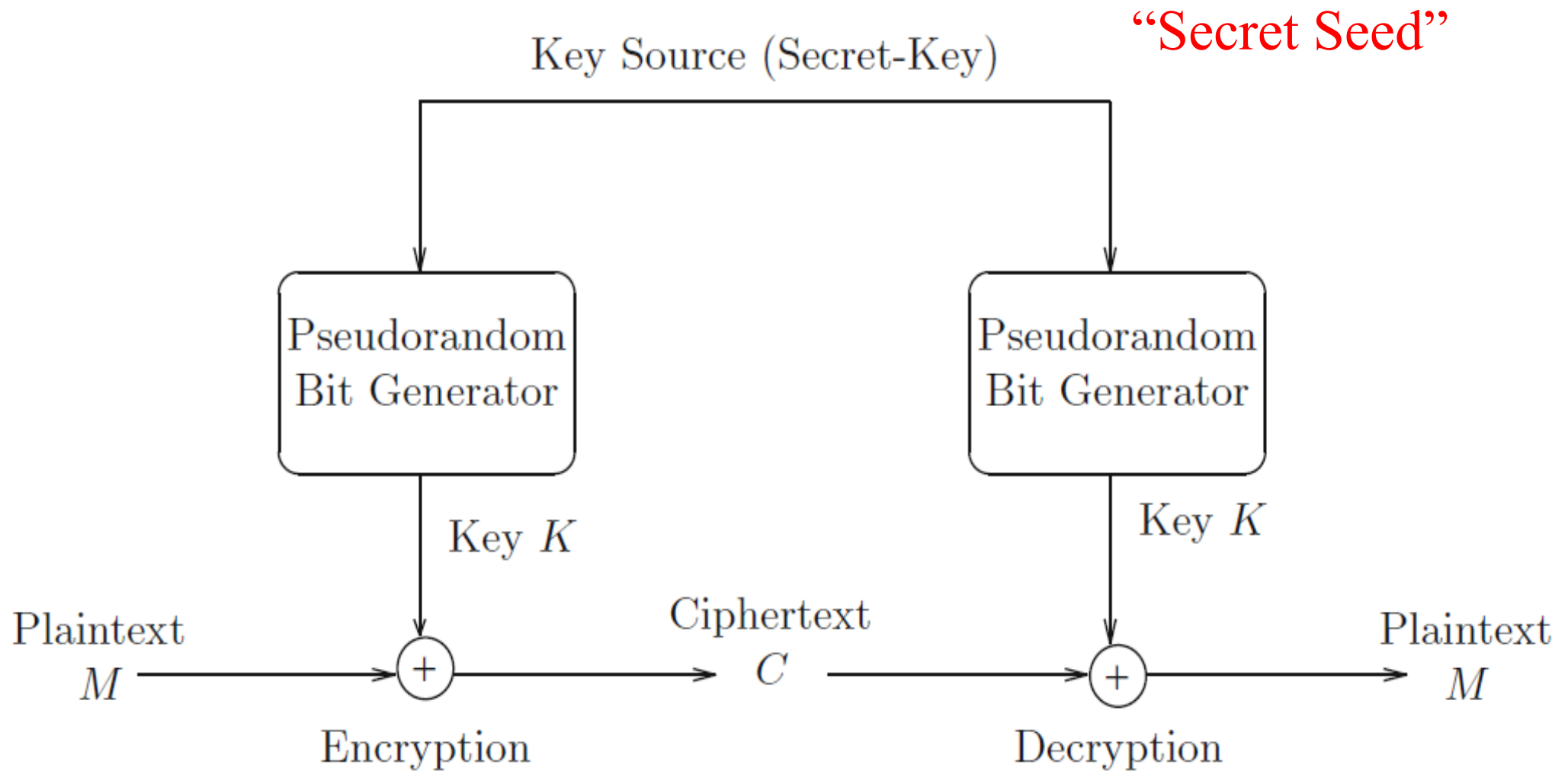
- Encrypts a digital data stream **one bit or one byte** at a time
  - Examples:
    - **Autokeyed** Vigenère cipher
    - Vernam cipher
- In the ideal case, a one-time pad version of the Vernam cipher would be used, in which the keystream is as long as the plaintext bit stream
  - If the cryptographic keystream is random, then this cipher is unbreakable by any means other than acquiring the keystream
    - Keystream must be provided to both users in advance via some independent and secure channel
    - This introduces insurmountable logistical problems if the intended data traffic is very large

# Stream Cipher (4 of 8)

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- For practical: must be implemented as an algorithmic to **generate key bit stream** (both users)
  - It must be computationally impractical to predict future portions of the bit stream based on previous portions of the bit stream
  - The two users need only share the **generating key** and each can produce the keystream

# Stream Cipher (5 of 8)



# Stream Cipher (6 of 8)

## ➤ Rivest Cipher 4

<https://en.wikipedia.org/wiki/RC4>

## ➤ Chaotic-based cryptosystem

[https://en.wikipedia.org/wiki/List\\_of\\_chaotic\\_maps](https://en.wikipedia.org/wiki/List_of_chaotic_maps)

| V · T · E           |                                                                                                                                                       | Stream ciphers                        |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Widely used ciphers | A5/1 · A5/2 · ChaCha · Crypto-1 · E0 · <b>RC4</b>                                                                                                     |                                       |
| eSTREAM Portfolio   | Software                                                                                                                                              | HC-256 · Rabbit · Salsa20 · SOSEMANUK |
|                     | Hardware                                                                                                                                              | Grain · MICKEY · Trivium              |
| Other ciphers       | Achterbahn · F-FCSR · FISH · ISAAC · MUGI · ORYX · Panama · Phelix · Pike · Py · QUAD · Scream · SEAL · SNOW · SOBER · SOBER-128 · VEST · VMPC · WAKE |                                       |
| Generators          | shrinking generator · self-shrinking generator · alternating step generator                                                                           |                                       |
| Theory              | block ciphers in stream mode · shift register · LFSR · NLFSR · T-function · IV                                                                        |                                       |
| Attacks             | correlation attack · correlation immunity · stream cipher attacks                                                                                     |                                       |

# Stream Cipher (7 of 8)

## ➤ Chaotic-based cryptosystem

Example:

Logistic map

$$x_{n+1} = rx_n(1 - x_n)$$

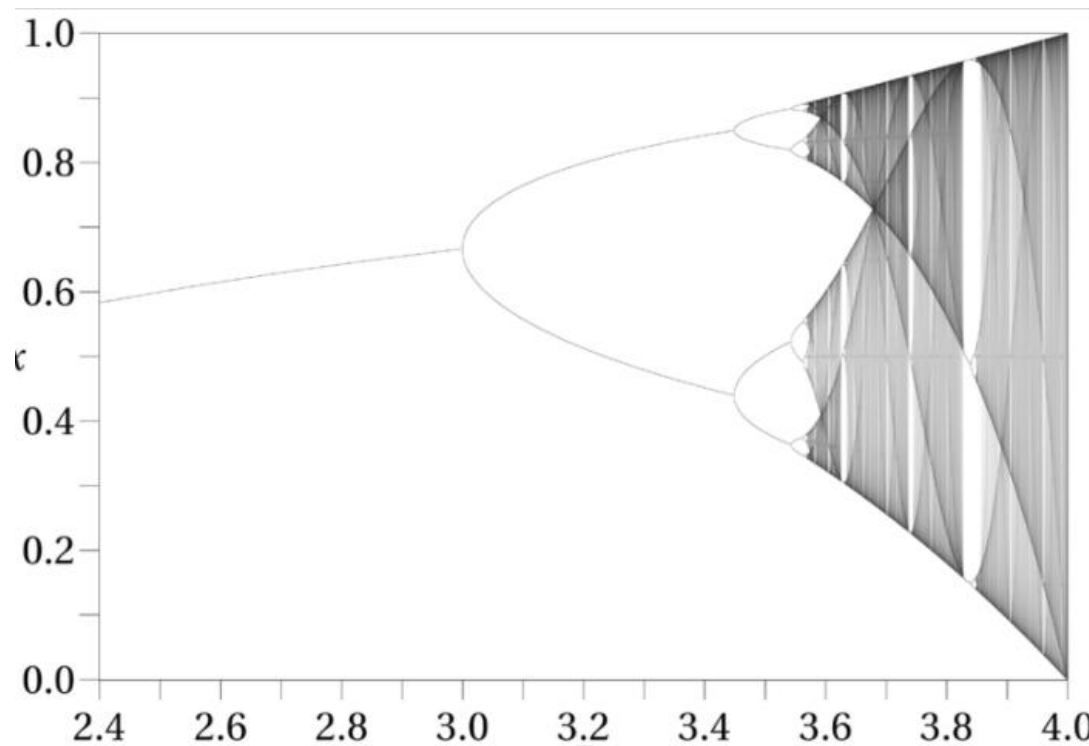
Input (seed):

$$x_0, r \in (3.6, 4)$$

Output:

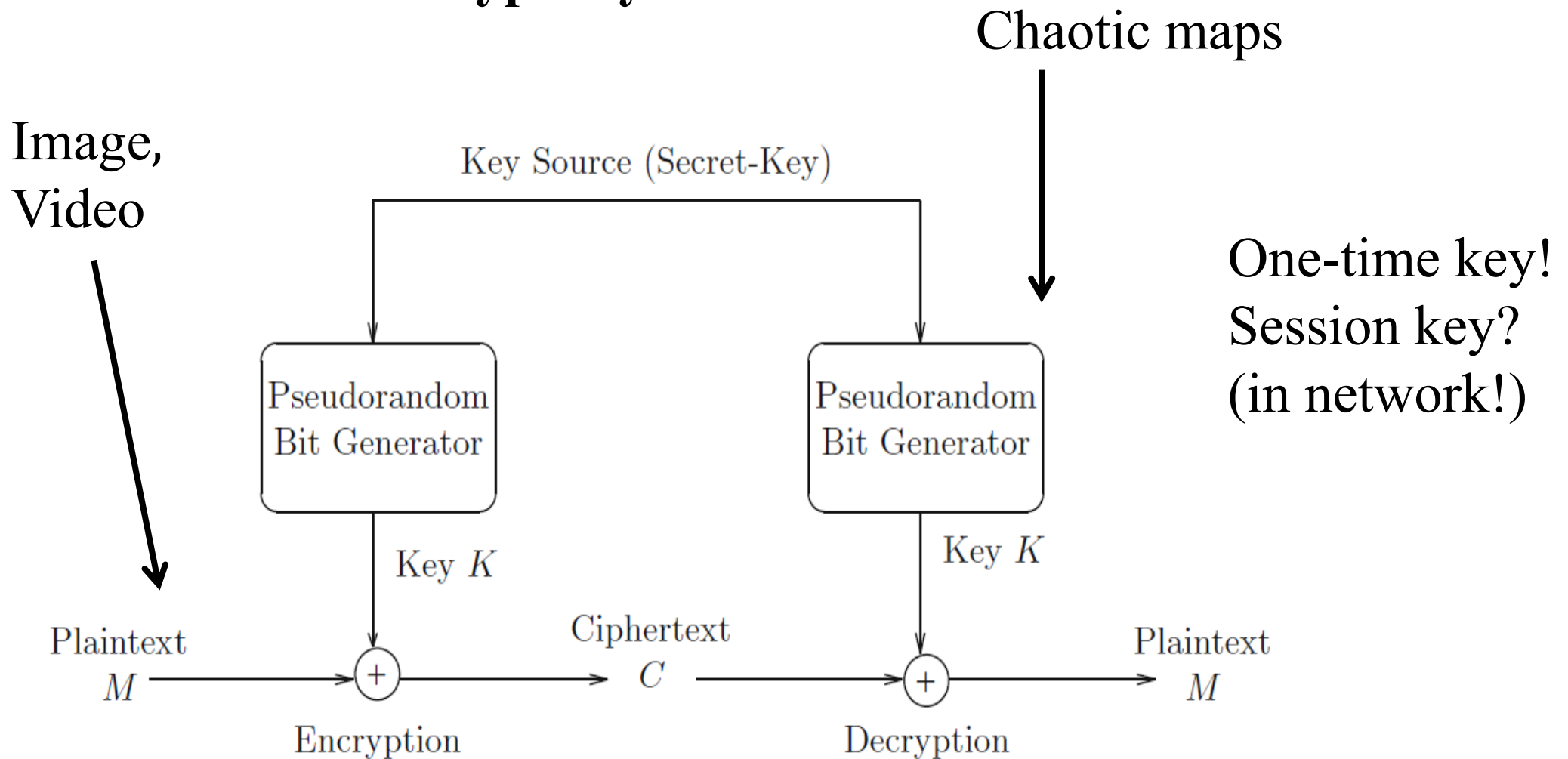
$$x_1, x_2, x_3, \dots x_n, \dots$$

$$0 < x_i < 1$$



# Stream Cipher (8 of 8)

## ➤ Chaotic-based crypto system



# Outline (week 4,5)

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- Stream Cipher
- Block cipher
  - Data Encryption Standard (DES)
  - Advanced Encryption Standard (AES)
  - Some other ciphers
    - Searchable encryption